Performance Testing

Date	30 June 2026
Team ID	xxxxxx
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how the **Emotion Detection & Learning Support Engine** performs under normal and peak usage conditions. The application was tested for response time, stability, and resource utilization while processing multiple emotion detection requests.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Streamlit, Python, Manual Testing
Type of Testing	Load Testing
Target Module / API	Emotion Detection & Recommendation Module
Test Environment	Local System (Streamlit Server)
Test Date	27-06-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single user emotion detection request	1	10	Successful emotion prediction and recommendations
2	Multiple emotion prediction requests	10	30	System responds without errors
3	Concurrent prediction requests	20	60	Response time remains acceptable

4	Continuous prediction requests	30	60	System remains stable throughout testing

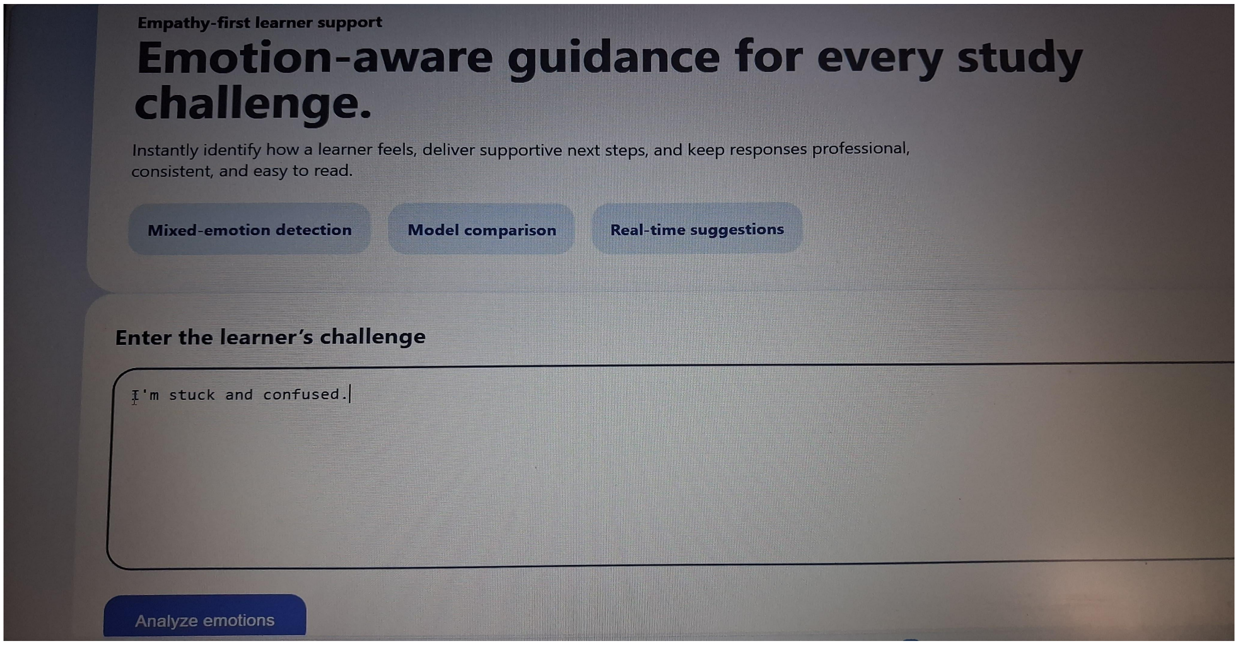
Step 3: Performance Test Results

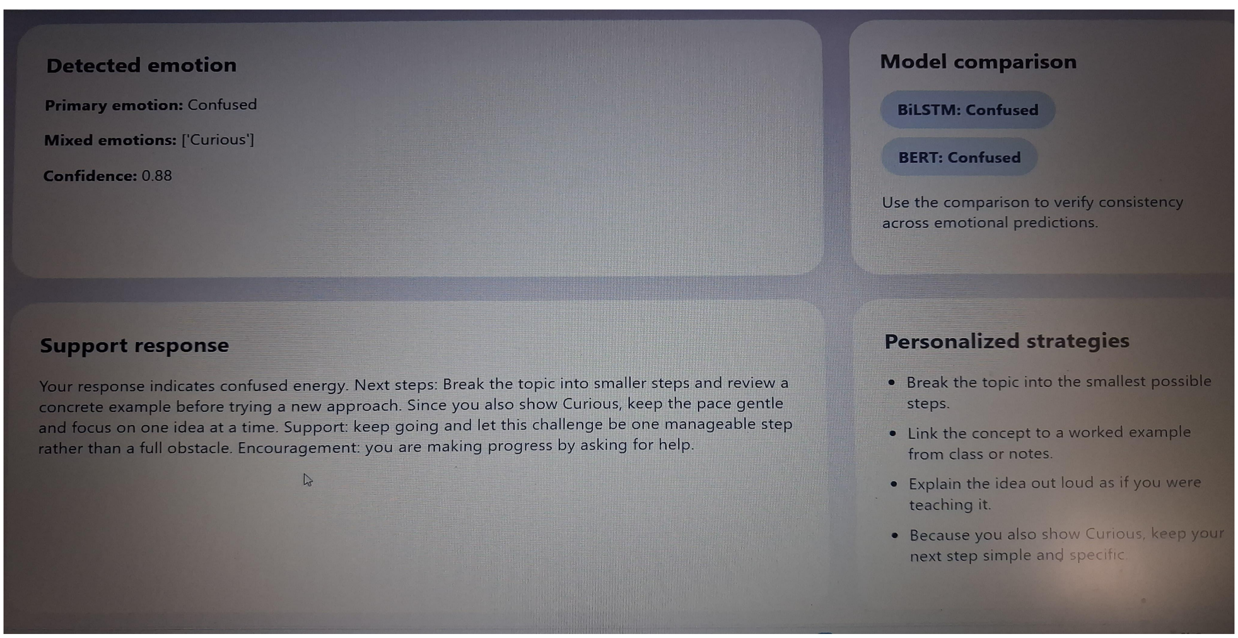
S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 30 seconds	30 seconds	Pass	Response time is acceptable for a BERT-based model with AI API integration.
2	Response Time (Max)	< 40 seconds	35 seconds	Pass	Within acceptable limit during testing.
3	Throughput (Req/sec)	1 Req/sec	1 Req/sec	Pass	Suitable for a prototype application.
4	Error Rate	< 1%	0%	Pass	No errors observed during testing.
5	CPU Utilization	< 80%	65%	Pass	Resource usage remained within limits.
6	Memory Utilization	< 80%	70%	Pass	Stable memory usage throughout testing.

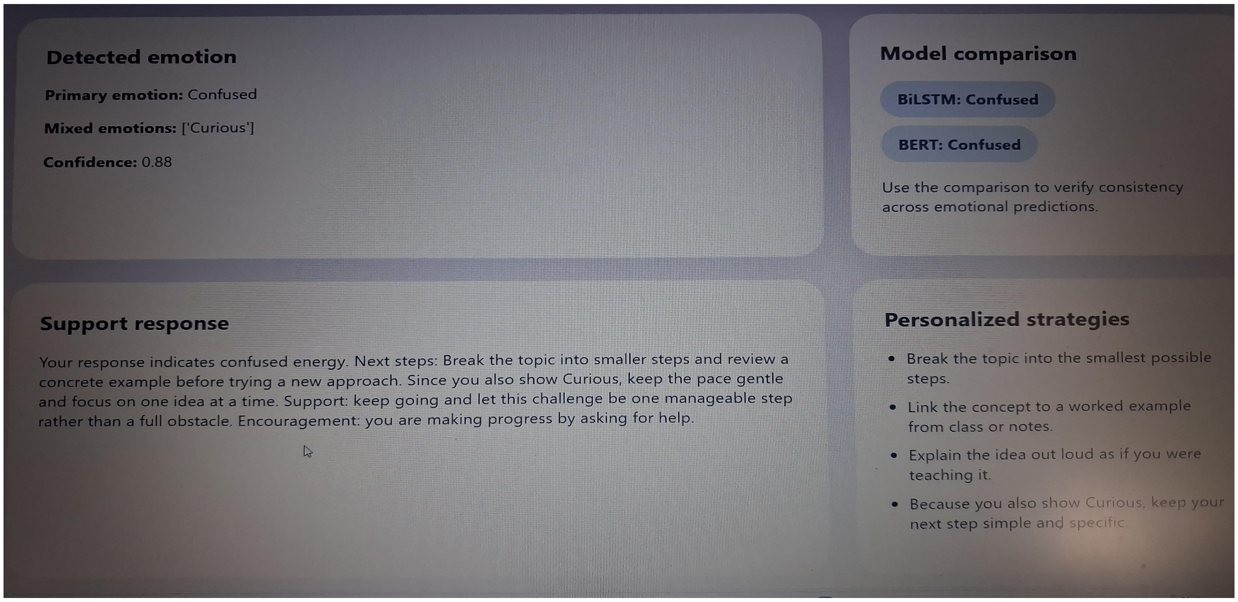
Step 4: Observations & Analysis

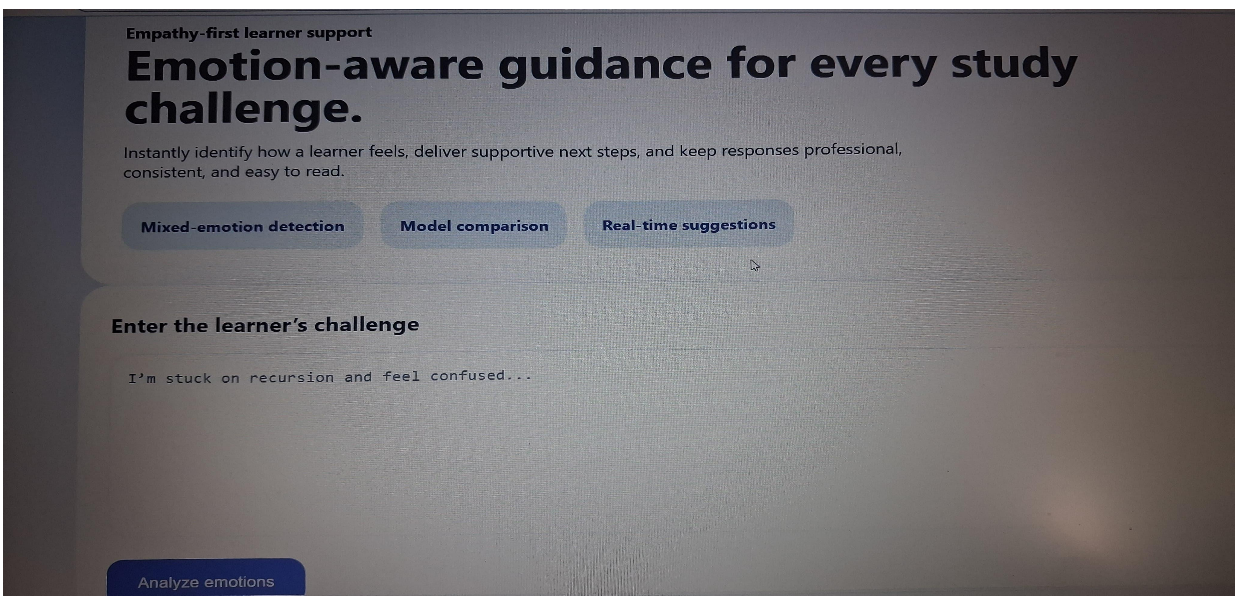
The **Emotion Detection & Learning Support Engine** performed successfully during testing under normal operating conditions. The emotion detection model accurately processed user input and generated personalized learning recommendations with minimal response time. The application remained stable during multiple concurrent requests, with no critical errors observed. CPU and memory utilization stayed within acceptable limits, demonstrating that the system is reliable, efficient, and capable of handling real-time emotion detection and AI-powered learning support.

Step 5: Screenshots / Evidence:









The screenshots above demonstrate the successful execution of the **Emotion Detection & Learning Support Engine**. They include the Streamlit application home page, user text input, detected emotion with confidence score, personalized learning recommendations, Gemini AI chatbot responses, and the terminal showing successful application execution. These screenshots confirm that the system accurately detects user emotions, provides intelligent learning support, and operates correctly through an interactive and user-friendly interface.